THE UNIVERSITY OF HONG KONG
DEPARTMENT OF MATHEMATICS
MATH2012 Fundamental Concepts of Mathematics
Solution to **Assignment 1**

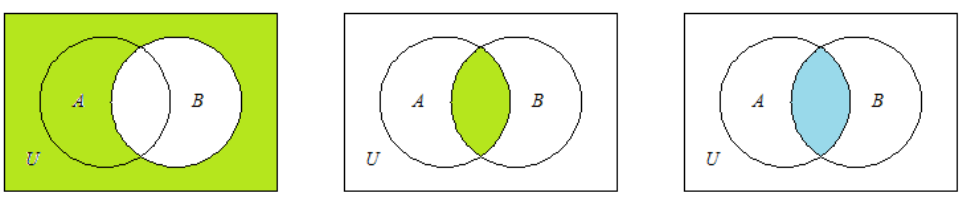
1. (a) $A = \{x \in \mathbb{N} : 2x^2+3x-2 = 0\} = \emptyset$ as the only two solutions to $2x^2+3x-2 = 0$ are $x = \frac{1}{2}$, -2 which are not natural numbers. Hence $|A| = 0$.

   (b) $B = \{x \in \mathbb{Z} : x^2 < 10\} = \{5\} = \{-3, -2, -1, 0, 1, 2, 3\}$ and hence $|B| = 7$.

   (c) $C = \{x \in \mathbb{R} : x^2+2x+3 = 0\} = \emptyset$ as there are no real solutions to $x^2+2x+3 = 0$. Hence $|C| = 0$.

   (d) Solving $|x-3| \geq 2$ gives $x \geq 5$ or $x \leq 1$. Thus the only integers in the first part of D that satisfy the inequalities are -1 and 6 only. In other words, $D = \{-1, 6\}$ and hence $|D| = 2$. Note that the elements $\{0, 1\}$, $\{5\}$, $\{7, 8\}$ of the first part of D are NOT integers and so they should not be included in the intersection.

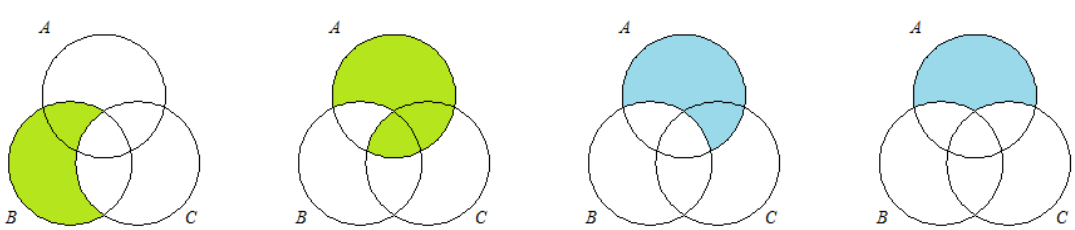
2. (a) We may consider the following Venn diagrams:



(a) for $U \setminus B$. (b) for $A \setminus (U \setminus B)$. (c) for $A \cap B$.

So $A \setminus (U \setminus B) = A \cap B$.

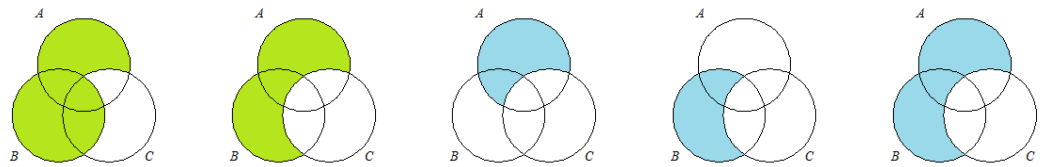
   (b) We may consider the following Venn diagrams:



(a) for $B \setminus C$. (b) for $A \setminus (B \setminus C)$. (c) for $A \setminus B$. (d) for $(A \setminus B) \setminus C$.

So $A \setminus (B \setminus C) \neq (A \setminus B) \setminus C$.

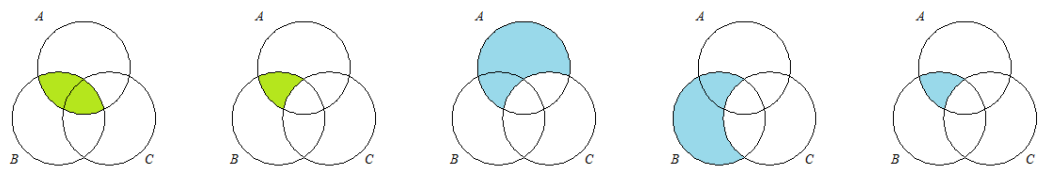
(c) We may consider the following Venn diagrams:



(a) for $A \cup B$. (b) for $(A \cup B) \setminus C$. (c) for $A \setminus C$. (d) for $B \setminus C$. (e) for $(A \setminus C) \cup (B \setminus C)$.

So $(A \cup B) \setminus C = (A \setminus C) \cup (B \setminus C)$.

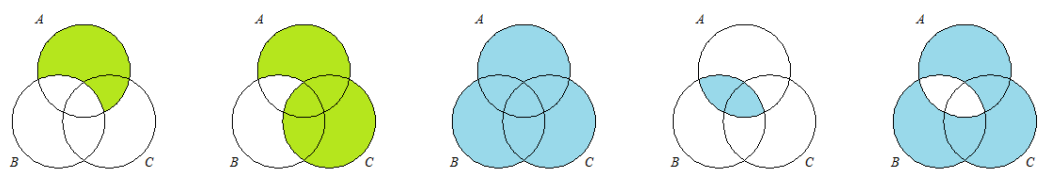
(d) We may consider the following Venn diagrams:



(a) for $A \cap B$. (b) for $(A \cap B) \setminus C$. (c) for $A \setminus C$. (d) for $B \setminus C$. (e) for $(A \setminus C) \cap (B \setminus C)$.

So $(A \cap B) \setminus C = (A \setminus C) \cap (B \setminus C)$.

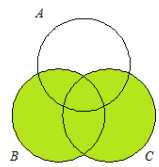
(e) We may consider the following Venn diagrams:



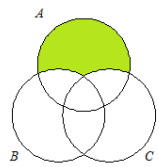
(a) for $A \setminus B$. (b) for $(A \setminus B) \cup C$. (c) for $A \cup B \cup C$. (d) for $A \cap B$. (e) for $(A \cup B \cup C) \setminus (A \cap B)$.

So $(A \setminus B) \cup C \neq (A \cup B \cup C) \setminus (A \cap B)$.

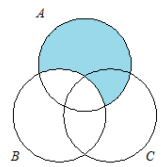
(f) We may consider the following Venn diagrams:



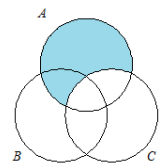
(a) for $B \cup C$.



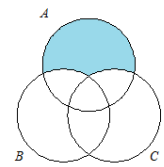
(b) for $A \setminus (B \cup C)$.



(c) for $A \setminus B$.



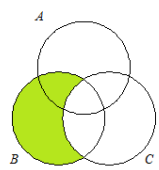
(d) for $A \setminus C$.



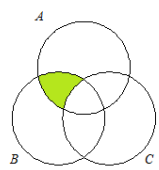
(e) for $(A \setminus B) \cap (A \setminus C)$.

So $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$.

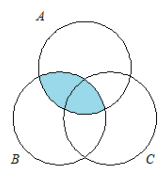
(g) We may consider the following Venn diagrams:



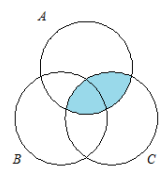
(a) for $B \setminus C$.



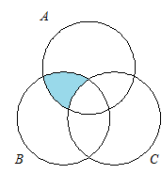
(b) for $A \cap (B \setminus C)$.



(c) for $A \cap B$.



(d) for $A \cap C$.



(e) for $(A \cap B) \setminus (A \cap C)$.

So $A \cap (B \setminus C) = (A \cap B) \setminus (A \cap C)$.

3. (a) $A = (0, 1) \cup (2, 3)$
   (b) $B = (-\infty, \frac{1}{2}) \cup (3, \infty)$

4. (a) $S = \{2x + 1 : x \in \mathbb{Z}\}$.
   (This is the set of all odd integers. You can also write S as $\{2x - 1 : x \in \mathbb{Z}\}$.)
   (b) $S = \{6x : x \in \mathbb{Z}\}$. (This is the set of all integers divisible by 6.)
   (c) $S = \{3x - 1 : x \in \mathbb{N}\}$ or $S = \{3x + 2 : x \in \mathbb{N} \cup \{0\}\}$.
   (d) $S = \{x^3 + 1 : x \in \mathbb{N}\}$.

5. (a) $\{b \in B : 2b \in A\} = \{1\}$.
   (b) $\{a \in A : a^2 \in B\} = \{1, 3\}$.
   (c) $\{(a, b) \in A \times B : a + b \text{ is odd}\}$
   $= \{(2, 1), (2, 3), (2, 5), (2, 7), (2, 9), (4, 1), (4, 3), (4, 5), (4, 7), (4, 9)\}$.

6. (a) **TRUE**. Empty set is a subset of ANY set.
   (b) **FALSE**. 1 is an element of $\mathbb{R}$ but 1 is not an element of $\{\mathbb{R}\}$.
   (c) **FALSE**. $\mathbb{R}$ is an element of $\{\mathbb{R}\}$ but $\mathbb{R}$ is not an element of $\mathbb{R}$.
   (d) **FALSE**. $\mathbb{Z}$ is an element of $\{\mathbb{Z}\}$ but $\mathbb{Z}$ is not an element of $\{\mathbb{R}\}$.

   <u>Remark</u>: In general, if we want to show that a set A is not a subset of a set B, we need to show that there is an element $x \in A$ but $x \notin B$.

7. (a) Note that $\mathcal{P}(A) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$ and $\mathcal{P}(B) = \{\emptyset, \{\emptyset\}\}$. So

$$\begin{aligned} & \mathcal{P}(A) \times \mathcal{P}(B) \\ = \; & \big\{(\emptyset, \emptyset), (\{a\}, \emptyset), (\{b\}, \emptyset), (\{a, b\}, \emptyset), (\emptyset, \{\emptyset\}), (\{a\}, \{\emptyset\}), (\{b\}, \{\emptyset\}), (\{a, b\}, \{\emptyset\})\big\}. \end{aligned}$$

   (b) $|\mathcal{P}(X \times Y)| = 2^{|X \times Y|} = 2^{|X| \cdot |Y|}$ and
   $|\mathcal{P}(X) \times \mathcal{P}(Y)| = |\mathcal{P}(X)| \cdot |\mathcal{P}(Y)| = 2^{|X|} \cdot 2^{|Y|} = 2^{|X|+|Y|}$.

8. For example, we can choose $A = \{\emptyset\}$, which is a nonempty set. Then $\mathcal{P}(A) = \{\emptyset, \{\emptyset\}\}$, and $A \subseteq \mathcal{P}(A)$ is true because every element of A (in this example $\emptyset$ is the only element of A) is an element of $\mathcal{P}(A)$.

9. (a) The set A is the set of those subsets S of $\mathbb{N}$ such that S is infinite. In other words, A is the set of all infinite subsets of $\mathbb{N}$. Thus each element of A is an infinite subset of $\mathbb{N}$. For example, since $\mathbb{N}$ itself is an infinite subset of $\mathbb{N}$, so $\mathbb{N}$ is an element of A.

   Another example of an element of A is given by the set of all positive even integers E as it is also an infinite subset of $\mathbb{N}$.

   (some other examples of elements of A: set of all positive multiples of 3; set of all positive multiples of 4; set of all prime numbers; $\ldots$, etc.)

   (b) The set B is the set of those subsets S of $\mathbb{N}$ such that S is infinite AND $\mathbb{N} \setminus S$ is infinite. In other words, B is the set of all infinite subsets of $\mathbb{N}$ such that their complements are also infinite subsets of $\mathbb{N}$. Similar to part (a), the set E of all positive even integers is an element of B because E itself is infinite and $\mathbb{N} \setminus E$, which is the set O of all positive odd integers, is also infinite.

   From this we can see that the set O gives another element of B with the same reason.

   However, unlike part (a), $\mathbb{N}$ is not an element of B because $\mathbb{N} \setminus \mathbb{N}$ is the empty set, which is not an infinite set.

   (some other examples of elements of B: set of all positive multiples of 3; set of all positive multiples of 4; set of all prime numbers; $\ldots$, etc.)

10. Based on the given information, we have

$$B = \{\{1, 2\}, \{1, 3\}, \{1, 4\}, \{2, 3\}, \{2, 4\}, \{3, 4\}\}.$$

Taking the intersection of any two distinct elements of the above set, we have

$$C = \{\emptyset, \{1\}, \{2\}, \{3\}, \{4\}\}.$$

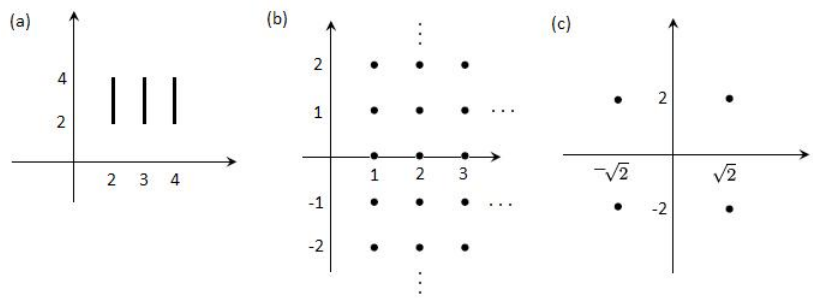
11. Since A and B are both finite, we can let $|A \setminus B| = m$ and $|B \setminus A| = n$ for some nonnegative integers m, n. As the two sets $A \setminus B$ and $B \setminus A$ are **disjoint**, we have

$$|A \triangle B| = |(A \setminus B) \cup (B \setminus A)| = |A \setminus B| + |B \setminus A| = m + n.$$

It follows that

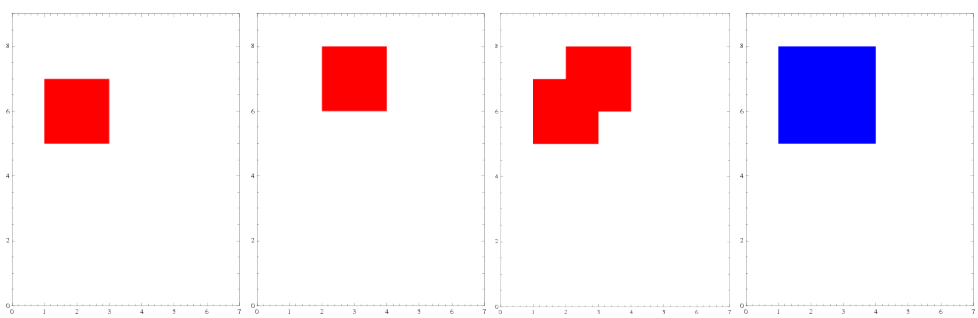
$$|\mathcal{P}(A \triangle B)| = 2^{m+n} = 2^m \cdot 2^n = |\mathcal{P}(A \setminus B)| \cdot |\mathcal{P}(B \setminus A)| = |\mathcal{P}(A \setminus B) \times \mathcal{P}(B \setminus A)|.$$

12. See the following figures:



13. Note that $A \cup B = [1, 4]$, $C \cup D = [5, 8]$, $A \cap B = [2, 3]$, $C \cap D = [6, 7]$.

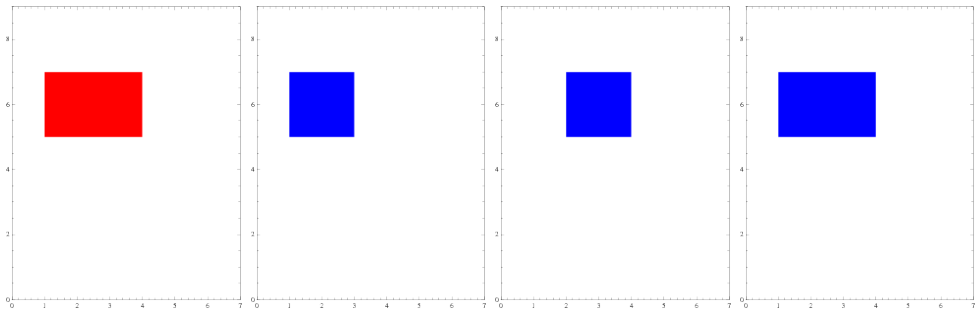
(a) Consider the following pictures,



(a) $A \times C$ (b) $B \times D$ (c) $(A \times C) \cup (B \times D)$ (d) $(A \cup B) \times (C \cup D)$

So $(A \times C) \cup (B \times D) \subseteq (A \cup B) \times (C \cup D)$.

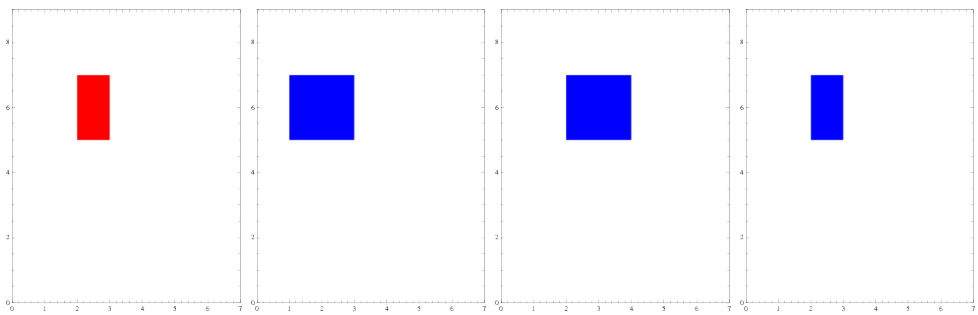
(b) Consider the following pictures,



(a) $(A \cup B) \times C$ (b) $A \times C$ (c) $B \times C$ (d) $(A \times C) \cup (B \times C)$

So $(A \cup B) \times C = (A \times C) \cup (B \times C)$.

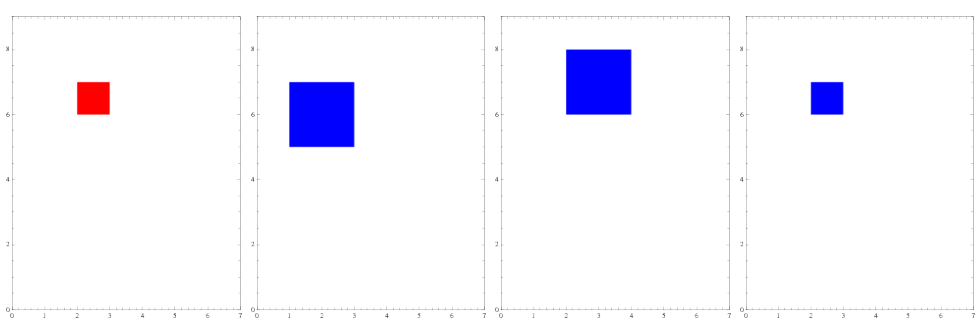
(c) Consider the following pictures,



(a) $(A \cap B) \times C$ (b) $(A \times C)$ (c) $B \times C$ (d) $(A \times C) \cap (B \times C)$

So $(A \cap B) \times C = (A \times C) \cap (B \times C)$.

(d) Consider the following pictures,



(a) $(A \cap B) \times (C \cap D)$ (b) $A \times C$ (c) $B \times D$ (d) $(A \times C) \cap (B \times D)$

So $(A \cap B) \times (C \cap D) = (A \times C) \cap (B \times D)$.

14. First of all, we have $(1, 3] \times \{2, 4\} = \{(a, b) : 1 < a \leq 3 \text{ and } (b = 2 \text{ or } 4)\}$. Then

$$\begin{aligned} S &= ((1, 3] \times \{2, 4\}) \cap \mathbb{Z}^2 \\ &= \{(a, b) : a \text{ and } b \text{ are integers and } 1 < a \leq 3 \text{ and } (b = 2 \text{ or } 4)\} \\ &= \{(2, 2), (2, 4), (3, 2), (3, 4)\}. \end{aligned}$$

15. $|B \times B| = 16$. For example,

$$((1,1),(1,2)) \in B \times B, \ ((2,1),(2,1)) \in B \times B.$$

16. (a) If $A = \{1\}$, then $\mathcal{P}(A) = \{\emptyset, \{1\}\}$. Thus we have

$$A \times \mathcal{P}(A) = \{(1,\emptyset),(1,\{1\})\}.$$

Then

$$B = \{(a,S) \in A \times \mathcal{P}(A) : a \in S\} = \{(1,\{1\})\}.$$

(b) If $A = \{1,2\}$, then $\mathcal{P}(A) = \{\emptyset, \{1\}, \{2\}, \{1,2\}\}$. Thus we have

$$A \times \mathcal{P}(A) = \{(1,\emptyset),(1,\{1\}),(1,\{2\}),(1,\{1,2\}),(2,\emptyset),(2,\{1\}),(2,\{2\}),(2,\{1,2\})\}.$$

Then

$$B = \{(a,S) \in A \times \mathcal{P}(A) : a \in S\} = \{(1,\{1\}),(1,\{1,2\}),(2,\{2\}),(2,\{1,2\})\}.$$

(c) If $A = \{1,2,3\}$, then $\mathcal{P}(A) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{2,3\}, \{1,2,3\}\}$. Thus we have

$$A \times \mathcal{P}(A) = \{(1,\emptyset),(1,\{1\}),(1,\{2\}),(1,\{3\}),\ldots,(3,\{2,3\}),(3,\{1,2,3\})\}.$$

Then

$$\begin{aligned} B &= \{(a,S) \in A \times \mathcal{P}(A) : a \in S\} \\ &= \{(1,\{1\}),(1,\{1,2\}),(1,\{1,3\}),(1,\{1,2,3\}),(2,\{2\}),(2,\{1,2\}),(2,\{2,3\}), \\ &\quad (2,\{1,2,3\}),(3,\{3\}),(3,\{1,3\}),(3,\{2,3\}),(3,\{1,2,3\})\} \end{aligned}$$

17. Let A, B and C be the set of students taking MATH1013, MATH2012 and MATH2241 this semester respectively. According to the given information, we have

$$|A \cup B \cup C| = 240, \quad |A| = 180, \quad |C| = 45, \quad |A \cap B| = 60, \quad |B \cap C| = 10$$
$$|A \cap C| = 0, \quad |A \cap B \cap C| = 0.$$

Then by *Theorem 1.38*, we have

$$240 = 180 + |B| + 45 - 60 - 10 - 0 + 0 \quad \Longrightarrow \quad |B| = 85.$$

18. Let $U = \{1,2,3,\ldots,1000\}$. Let

$$\begin{aligned} A &= \{x \in U : x \text{ is a multiple of } 2\}, & B &= \{x \in U : x \text{ is a multiple of } 3\}, \\ C &= \{x \in U : x \text{ is a multiple of } 5\}, & D &= \{x \in U : x \text{ is a multiple of } 7\}. \end{aligned}$$

We need to find $|A \cup B \cup C \cup D|$. According to the Inclusion-Exclusion Principle (*Theorem 1.41*), when $n = 4$, we have

$$\begin{aligned} |A \cup B \cup C \cup D| &= |A| + |B| + |C| + |D| - |A \cap B| - |A \cap C| \\ &\quad -|A \cap D| - |B \cap C| - |B \cap D| - |C \cap D| \\ &\quad +|A \cap B \cap C| + |A \cap B \cap D| + |A \cap C \cap D| + |B \cap C \cap D| \\ &\quad -|A \cap B \cap C \cap D|. \end{aligned}$$

It remains to find each term of the right hand side. It is easy to see that $|A| = 500$ and $|C| = 200$. For $|B|$, we note that the number of integers from 1 to 1000 that are multiples of 3 will be the greatest integer less than or equal to $\frac{1000}{3}$, that is, $|B| = 333$. Similar consideration gives $|D| = 142$.

Next, an integer is in $A \cap B$ if and only if it is a multiple of both 2 and 3. Thus $|A \cap B|$ will be the greatest integer less than or equal to $\frac{1000}{6}$, that is, $|A \cap B| = 166$. Similarly, we have

$$|A \cap C| = 100, \quad |A \cap D| = 71, \quad |B \cap C| = 66, \quad |B \cap D| = 47, \quad |C \cap D| = 28,$$

$$|A \cap B \cap C| = 33, \quad |A \cap B \cap D| = 23, \quad |A \cap C \cap D| = 14, \quad |B \cap C \cap D| = 9$$

and $|A \cap B \cap C \cap D| = 4$. Hence

$$\begin{aligned} |A \cup B \cup C \cup D| &= 500 + 333 + 200 + 142 - 166 - 100 - 71 - 66 - 47 - 28 \\ &\quad + 33 + 23 + 14 + 9 - 4 \\ &= 772. \end{aligned}$$